

DS 200

Team Project 1

Nov 9, 2023

RDBMS Implementation Project

Team Members:

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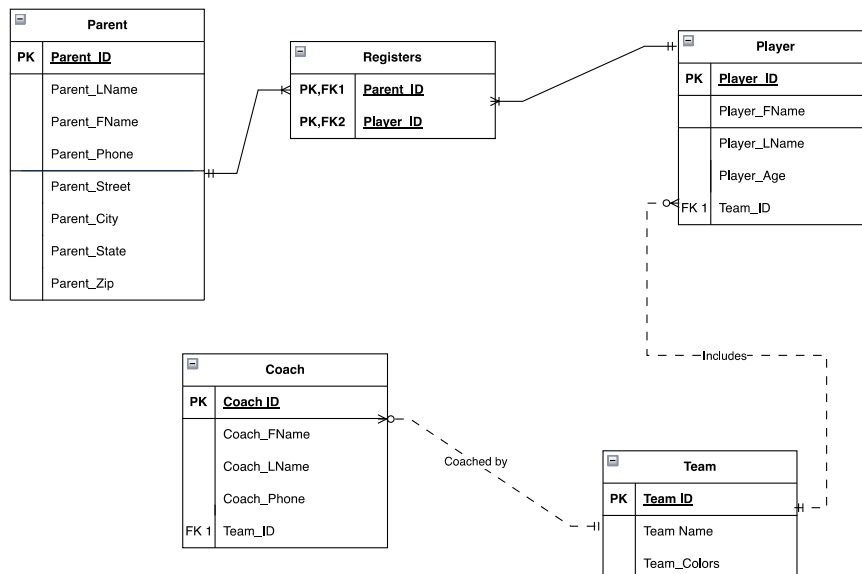
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Shwetha Arumugam Chitra Devi: Queries Developer

Shen-Han Chiu: EER Analyst

Conceptual Data Model



SQL Scripts for Our Database Tables and Implementation

```
1 CREATE TABLE Team(  
2   TeamID INT,  
3   TeamName VARCHAR,  
4   TeamColor VARCHAR,  
5   PRIMARY KEY("TeamID")  
6 );  
7  
8 CREATE TABLE Player(  
9   PlayerID INT,  
10  TeamID INT,  
11  PlayerFirstName VARCHAR,  
12  PlayerLastName VARCHAR,  
13  PlayerAge INT,  
14  PRIMARY KEY("PlayerID"),  
15  FOREIGN KEY("TeamID") REFERENCES Team("TeamID")  
16 );  
17  
18 CREATE TABLE Coach(  
19   CoachID INT,  
20   TeamID INT,  
21   CoachFirstName VARCHAR,  
22   CoachLastName VARCHAR,  
23   CoachHomePhoneNumber INT,  
24   PRIMARY KEY("CoachID")  
25   FOREIGN KEY("TeamID") REFERENCES Team("TeamID")  
26 );  
27  
28 CREATE TABLE Parent (  
29   ParentID INT,  
30   ParentFirstName VARCHAR,  
31   ParentLastName VARCHAR,  
32   HomePhoneNumber INT,  
33   Street VARCHAR,  
34   City VARCHAR,  
35   State VARCHAR,  
36   ZipCode VARCHAR,  
37   PRIMARY KEY("ParentID")  
38 );  
39  
40 CREATE TABLE Register (  
41   PlayerID INT,  
42   ParentID INT,  
43   FOREIGN KEY(ParentID) REFERENCES Parent(ParentID),  
44   FOREIGN KEY(PlayerID) REFERENCES Player(PlayerID),  
45   PRIMARY KEY(PlayerID,ParentID)  
46 );
```

Execution finished without errors.
Result: query executed successfully. Took 0ms

Team Table:

TeamID: Unique team identifier.

TeamName: Team's name.

TeamColors: Colors associated with the team.

Player Table:

PlayerID: Unique player identifier.
TeamID: Foreign key linking to the Team table.
PlayerFirstName, PlayerLastName: Player's name.
PlayerAge: Player's age.

Coach Table:

CoachID: Unique coach identifier.
TeamID: Foreign key linking to the Team table.
CoachFirstName, CoachLastName: Coach's name.
CoachHomePhoneNumber: Coach's home phone number.

Parent Table:

ParentID: Unique parent identifier.
ParentLastName, ParentFirstName: Parent's name.
HomePhoneNumber: Parent's home phone number.
Street: Parent's street address.
City: Parent's City.
State: Parent's State.
ZipCode: Parent's Zipcode

Register Table:

PlayerID, ParentID: Composite primary key linking players to parents.
Foreign keys linking to Player and Parent tables, ensuring relationships between players and their parents.

This SQLite schema establishes tables for a soccer league database. The 'Team' table records team information, while 'Player' and 'Coach' tables link to teams. 'Parent' contains details of parents. The 'PlayerParent' table associates players with multiple parents. Foreign key constraints ensure data accuracy and enable seamless queries for comprehensive league management.

Data Implementation Result

1. Team

	TeamID	TeamName	TeamColor
	Filter	Filter	Filter
1	1	Manches...	Red
2	2	Chelsea ...	Blue
3	3	Real ...	White
4	4	FC ...	Blue
5	5	Liverpool...	Red

2. Player

	PlayerID	TeamID	PlayerFirstName	PlayerLastName	PlayerAge
	Filter	Filter	Filter	Filter	Filter
1	1	1	David	Beckham	15
2	2	1	Wayne	Rooney	15
3	3	2	Frank	Lampard	12
4	4	3	Cristiano	Ronaldo	16
5	5	4	Lionel	Messi	14

3. Coach

	CoachID	TeamID	CoachFirstName	CoachLastName	CoachHomePhoneNumber
	Filter	Filter	Filter	Filter	Filter
1	1	1	Ole	Gunnar Solskjær	123-456-7890
2	2	2	Thomas	Tuchel	987-654-3210
3	3	3	Zinedine	Zidane	555-123-4567
4	4	4	Ronald	Koeman	777-987-6543
5	5	5	Jurgen	Klopp	111-222-3333

4. Parent

	ParentID	ParentFirstName	ParentLastName	HomePhoneNumber	Street	City	State	ipCod
	Filter	Filter	Filter	Filter	Filter	Filter	Filt...	Filter
1	1	John	Bekham	555-123-4567	123 Main St	Any...	CA	90...
2	2	Alice	Rooney	777-987-6543	456 Elm St	Oth...	NY	10...
3	3	Robert	Lampard	111-222-3333	789 Oak St	Sm...	TX	75...
4	4	Laura	Ronaldo	444-555-6666	321 Maple St	Ne...	IL	60...
5	5	Michael	Messi	999-888-7777	654 Pine St	Bigt...	FL	33...

5. Register

	PlayerID	ParentID
	Filter	Filter
1	1	1
2	2	1
3	3	2
4	4	3
5	5	4

Soccer teams (Manchester United, Chelsea FC, Real Madrid, FC Barcelona, Liverpool FC) are established, each with a unique identity and team color.

Players (David Beckham, Wayne Rooney, Frank Lampard, Cristiano Ronaldo, Lionel Messi) are associated with specific teams based on their IDs, with additional details such as age provided.

Coaches (Ole Gunnar Solskjær, Thomas Tuchel, Zinedine Zidane, Ronald Koeman, Jurgen Klopp) are assigned to teams along with their contact numbers.

Parents (John Bekham, Alice Rooney, Robert Lampard, Laura Ronaldo, Michael Messi) are documented with contact details and home addresses.

The **Register** table establishes connections between players and their parents, signifying relationships such as David Beckham being associated with John Bekham (PlayerID 1, ParentID 1).

Test Runs of Implementation that Answer the 6 Typical Queries

1. get all players for specific team

here I have used specific_team_id but it is whatever the user chooses:

Code:

```
SELECT PlayerFirstName, PlayerLastName
FROM Player
WHERE TeamID = [specific_team_id];
```

```
1  SELECT PlayerFirstName, PlayerLastName
2  FROM Player
3  WHERE TeamID = 1;
4
5
```

	PlayerFirstName	PlayerLastName
1	David	Beckham
2	Wayne	Rooney

```
Execution finished without errors.
Result: 2 rows returned in 3ms
At line 1:
SELECT PlayerFirstName, PlayerLastName
FROM Player
WHERE TeamID = 1;
```

2. get coaches for a specific team:

here I have used specific_team_id but it is whatever the user chooses:

Code:

```
SELECT CoachFirstName, CoachLastName  
FROM Coach  
WHERE TeamID = [specific_team_id];
```

```
1  SELECT CoachFirstName, CoachLastName  
2  FROM Coach  
3  WHERE TeamID = 3;  
4
```

	CoachFirstName	CoachLastName
1	Zinedine	Zidane

```
Execution finished without errors.  
Result: 1 rows returned in 4ms  
At line 1:  
SELECT CoachFirstName, CoachLastName  
FROM Coach  
WHERE TeamID = 3;
```

3. get parents of a specific player:

here I have used specific_player_id but it is whatever the user chooses:

Code:

```
SELECT ParentFirstName, ParentLastName  
FROM Parent  
JOIN Register ON Parent.ParentID = Register.ParentID  
WHERE Register.PlayerID= [specific_player_id];
```

```

1  SELECT ParentFirstName, ParentLastName
2  FROM Parent
3  JOIN Register ON Parent.ParentID = Register.ParentID
4  WHERE Register.PlayerID= 2;
5
6

```

	ParentFirstName	ParentLastName
1	John	Bekham

Execution finished without errors.
 Result: 1 rows returned in 3ms
 At line 1:
 SELECT ParentFirstName, ParentLastName
 FROM Parent
 JOIN Register ON Parent.ParentID = Register.ParentID
 WHERE Register.PlayerID= 2;

4. get all players whose parents live in a specific city:

```

SELECT PlayerFirstName, PlayerLastName
FROM Player
JOIN Register ON Player.PlayerID = Register.PlayerID
JOIN Parent ON Register.ParentID = Parent.ParentID
WHERE Parent.City = '[specific_city]' ;

```

```

1  SELECT PlayerFirstName, PlayerLastName
2  FROM Player
3  JOIN Register ON Player.PlayerID = Register.PlayerID
4  JOIN Parent ON Register.ParentID = Parent.ParentID
5  WHERE Parent.City = 'Smalltown' ;
6
7

```

	PlayerFirstName	PlayerLastName
1	Cristiano	Ronaldo

Execution finished without errors.
 Result: 1 rows returned in 3ms
 At line 1:
 SELECT PlayerFirstName, PlayerLastName
 FROM Player
 JOIN Register ON Player.PlayerID = Register.PlayerID
 JOIN Parent ON Register.ParentID = Parent.ParentID
 WHERE Parent.City = 'Smalltown' ;

5. get the team information for a specific coach:

```

SELECT Team.TeamName, Team.TeamColor
FROM Team
WHERE Team.TeamID = (SELECT Coach.TeamID FROM Coach WHERE
Coach.CoachID = [specific_coach])

```

```

1  SELECT Team.TeamName, Team.TeamColor
2  FROM Team
3  WHERE Team.TeamID = (SELECT Coach.TeamID FROM Coach WHERE Coach.CoachID = 4);
4
5
6

```

	TeamName	TeamColor
1	FC Barcelona	Blue

Execution finished without errors.
Result: 1 rows returned in 10ms
At line 1:
SELECT Team.TeamName, Team.TeamColor
FROM Team
WHERE Team.TeamID = (SELECT Coach.TeamID FROM Coach WHERE Coach.CoachID = 4);

6. get all players who are above a specific age:

```

SELECT PlayerFirstName, PlayerLastName
FROM Player
WHERE PlayerAge > [specific_age];

```

```

1  SELECT PlayerFirstName, PlayerLastName
2  FROM Player
3  WHERE PlayerAge > 13;
4
5
6
7

```

	PlayerFirstName	PlayerLastName
1	David	Beckham
2	Wayne	Rooney
3	Cristiano	Ronaldo
4	Lionel	Messi

Execution finished without errors.
Result: 4 rows returned in 15ms
At line 1:
SELECT PlayerFirstName, PlayerLastName
FROM Player
WHERE PlayerAge > 13;